

A Formal C2 Route to Off-Critical Nonvanishing for the Riemann Zeta Function

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Abstract

We introduce a dyadic carry framework C2 and define the genuine analytic numerator $F(s) = D_\infty(s) - B_\infty(s)$. In the half-plane of absolute convergence the C2 bracket bijection gives an exact lateral cancellation, and the remaining central tower yields the identity $F(s) = c_0(s)\zeta(s)$, with an explicit nonzero normalizing factor c_0 . The C2 geometry has an internal barrier at $\text{Re}(s) = 1/2$: the branch norm depends only on $\sigma = \text{Re}(s)$, while the transversal tilt detects displacement from the critical line. The proof route combines near-critical-zero rigidity, middle-region resolvent non-cancellation, cutoff/residual control, and a regional gluing argument. The accompanying Lean development exposes the audited route through `LeanC2.PeerReview`; its terminal public theorem is the certified implication displayed in Theorem 1.1, from a terminal C2 certificate to mathlib’s official `RiemannHypothesis` statement. This draft gives the human mathematical route and records the exact Lean correspondence for external audit.

1 Introduction

The Riemann hypothesis may be formulated as a nonvanishing statement for $\zeta(s)$ away from the critical line inside the critical strip. The C2 development studied here approaches that problem through a genuine dyadic numerator

$$F(s) := D_\infty(s) - B_\infty(s),$$

a formal identity relating F to ζ , and a certificate-driven proof of nonvanishing for F on the off-critical strip.

The route has the following form:

$$\begin{aligned} & \text{C2 bracket geometry} \longrightarrow \text{genuine numerator } F \longrightarrow F = c_0\zeta \\ & \longrightarrow \text{near/bulk/edge certificates} \longrightarrow \text{off-critical regional cover} \longrightarrow F(s) \neq 0 \text{ off-axis} \\ & \longrightarrow \text{C2/zeta transfer} \longrightarrow \text{off-critical nonvanishing of } \zeta \longrightarrow \text{RiemannHypothesis}. \end{aligned}$$

The main design choice is to separate the analytic obligations from the transport and assembly lemmas. In the Lean development, the terminal theorem is not stated as an informal collection of hidden hypotheses. Instead, it consumes an inhabitant of the certificate type:

`C2.PeerReview.RiemannHypothesisTerminalCertificate.`

The theorem

`C2.PeerReview.RiemannHypothesis_of_terminalCertificate`

then returns mathlib’s `RiemannHypothesis` proposition.

1.1 Main statements

Theorem 1.1 (Lean-certified terminal implication). *In the review-facing Lean facade, the following theorem is checked:*

```
C2.PeerReview.RiemannHypothesis_of_terminalCertificate :  
  C2.PeerReview.RiemannHypothesisTerminalCertificate ->  
  RiemannHypothesis
```

For audit clarity, the current review facade does not expose a closed zero-argument constant named `C2.PeerReview.terminalCertificate`. The formal claim recorded in this draft is therefore the certified implication from the terminal certificate interface, together with the public inventory of the fields and obligations needed to inhabit that interface. The same facade also exposes the canonical assembly layer

```
C2.PeerReview.TerminalCertificate_of_continuedModelMiddleLocal  
C2.PeerReview.RiemannHypothesis_of_continuedModelMiddleLocal
```

which builds the terminal package and derives `RiemannHypothesis` from the canonical near-axis certificate for the continued C2 model together with the canonical middle-local witness. In addition, the facade exposes sharper endpoints where the middle-local witness is replaced by pointwise horizontal-budget and residual-dominance hypotheses:

```
C2.PeerReview.TerminalCertificate_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds  
C2.PeerReview.RiemannHypothesis_of_continuedModelCanonicalClosedScaledMiddlePointwiseBounds  
C2.PeerReview.RiemannHypothesis_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds  
C2.PeerReview.RiemannHypothesis_of_continuedModelCanonicalClosedScaledMiddleResidualAnalyticBounds
```

The continuation-based constructor remains available as an alternate audit interface, but the current terminal assembly no longer needs it for the near-axis package. Thus the terminal certificate is an audit boundary for the current route, not a hidden or informal premise.

Theorem 1.2 (C2 mathematical route, informal). *The intended mathematical route is to prove that the genuine C2 numerator $F = D_\infty - B_\infty$ is nonvanishing on the off-critical strip. Since $F = c_0\zeta$ and c_0 is nonzero there, this transfers to off-critical nonvanishing of ζ , and hence to mathlib’s `RiemannHypothesis` statement.*

1.2 Contributions

This paper records the following review-facing contributions.

- (i) A dyadic bracket decomposition for odd integers and a formal lateral cancellation theorem for $D_\infty - B_\infty$.
- (ii) A genuine-first identity $F = c_0\zeta$, with $F = D_\infty - B_\infty$ and explicit c_0 .
- (iii) A formal C2/zeta transfer layer from C2 operator nonvanishing to off-critical nonvanishing of ζ , and hence to mathlib’s `RiemannHypothesis`.
- (iv) A regional-cover abstraction for the off-critical strip.
- (v) A continuation package for the genuine C2 operator and its near-critical-zero certificate.
- (vi) A middle-region nonvanishing interface based on resolvent non-cancellation: a dominant four-level block is compared against explicit residual budgets.
- (vii) A terminal certificate interface and checked Lean endpoint wrapper that assemble the route once the certificate fields are supplied.

1.3 Audit boundary

The formalization deliberately distinguishes three kinds of statements.

- (1) Conceptual proof architecture: identity, cover, near-critical-zero nonvanishing, middle-region non-cancellation, edge handling, and transfer.
- (2) Certificate data: bound packages, residual budgets, local witnesses, oscillatory witnesses, and cover data.
- (3) Lean wrappers and packaging lemmas: aliases and theorem wrappers in the review-facing facade `LeanC2.PeerReview`.

The central endpoint theorem is therefore best read as a checked contract: given the terminal C2 certificate, Lean derives the official `RiemannHypothesis` proposition. The certificate inventory is public and is documented in `docs/BOUNDS_CERTIFICATES_WITNESSES.md`.

2 The C2 dyadic decomposition

The elementary C2 address system is defined by the two-adic valuation

$$v_2(n) = \max\{k : 2^k \mid n\}.$$

For an odd integer $n \geq 3$, its effective C2 depth is

$$k_{\text{eff}}(n) = \max(v_2(n-1), v_2(n+1)).$$

A C2 center has the form

$$c = 2^k m, \quad k \geq 2, \quad m \text{ odd},$$

and its two odd legs are $c-1$ and $c+1$.

Theorem 2.1 (C2 bracket bijection). *Every odd integer $n \geq 3$ appears as exactly one leg of exactly one C2 bracket. More precisely, if $c = 2^k m$ with $k \geq 2$ and m odd, then*

$$k_{\text{eff}}(c-1) = k_{\text{eff}}(c+1) = k.$$

Conversely, every odd $n \geq 3$ is either $2^k m - 1$ or $2^k m + 1$ for a unique C2 address $(k, m, \pm)$ with $k \geq 2$ and m odd.

This section is purely combinatorial. Its purpose is to show that the odd integers can be regrouped by dyadic brackets without loss or multiplicity. In Lean, the existence and uniqueness statements are formalized as

`C2.bracket_bijection_odd_ge_three_exists,`
`C2.bracket_bijection_odd_ge_three.`

The corresponding weight identity is

`C2.halving_law_odd_ge_three.`

3 The genuine C2 numerator

The genuine direct lateral operator is

$$D_{\infty}(s) = \sum_{\substack{n \text{ odd} \\ n \geq 3}} 2^{-k_{\text{eff}}(n)} n^{-s}.$$

The genuine bracket operator is

$$B_{\infty}(s) = \sum_{k \geq 2} \sum_{m \text{ odd}} 2^{-k} ((c-1)^{-s} + (c+1)^{-s} - 2c^{-s}), \quad c = 2^k m.$$

We define the genuine C2 numerator by

$$F(s) := D_{\infty}(s) - B_{\infty}(s).$$

The bracket bijection implies that the lateral terms in D_{∞} and B_{∞} match exactly. Hence, in the half-plane where the series are absolutely convergent,

$$D_{\infty}(s) - B_{\infty}(s) = 2 \sum_{k \geq 2} \sum_{m \text{ odd}} 2^{-k} (2^k m)^{-s}.$$

This cancellation is exact. It is not a numerical observation and does not depend on a cutoff.

The corresponding Lean declarations are:

```
C2.directLateralInfinite_sub_bracketLateralInfinite_eq_central_of_one_lt_re
C2.genuineFInfinite_eq_central_of_one_lt_re
```

4 Formal setup

Let $s = \sigma + it \in \mathbb{C}$. The critical strip is the region

$$0 < \text{Re}(s) < 1,$$

and the critical line is $\text{Re}(s) = 1/2$. The off-critical strip is the portion of the critical strip with $\text{Re}(s) \neq 1/2$.

Definition 4.1 (Off-critical nonvanishing). *For a function $F : \mathbb{C} \rightarrow \mathbb{C}$, off-critical nonvanishing means*

$$F(s) \neq 0$$

for every s in the off-critical strip.

In the public Lean facade this predicate is exposed as

```
C2.PeerReview.OffCriticalNonvanishing.
```

The official endpoint is not a local replacement for the Riemann hypothesis. It is the proposition already provided by mathlib:

```
RiemannHypothesis.
```

5 The fundamental identity $F = c_0\zeta$

From the central form of the genuine numerator,

$$F(s) = 2 \sum_{k \geq 2} \sum_{m \text{ odd}} 2^{-k} (2^k m)^{-s},$$

we separate the depth and odd-core variables:

$$F(s) = 2 \left(\sum_{k \geq 2} 2^{-k(1+s)} \right) \left(\sum_{m \text{ odd}} m^{-s} \right).$$

The odd Dirichlet channel is

$$\sum_{m \text{ odd}} m^{-s} = (1 - 2^{-s})\zeta(s),$$

and the depth sum is

$$\sum_{k \geq 2} 2^{-k(1+s)} = \frac{2^{-2(1+s)}}{1 - 2^{-(1+s)}}.$$

Thus

$$F(s) = c_0(s)\zeta(s), \quad c_0(s) = \frac{2^{-2s}(2^s - 1)}{2 \cdot 2^s - 1}.$$

In the Lean source, the identity in the absolute-convergence half-plane is checked by

`C2.genuineFInfinite_eq_c0_mul_riemannZeta_of_one_lt_re.`

The public route consumes the corresponding continuation data through

`C2.PeerReview.GenuineC2ContinuationData.`

5.1 Nonvanishing of the normalizing factor

On the critical line $s = 1/2 + it$, writing $2^s = \sqrt{2} e^{it \log 2}$ gives

$$c_0(s) = \frac{2^{-2s}(2^s - 1)}{2 \cdot 2^s - 1}.$$

Neither $2^s - 1$ nor $2 \cdot 2^s - 1$ can vanish on that line: their equations would force a unit complex number to have modulus $1/\sqrt{2}$ or $1/(2\sqrt{2})$. More generally, in the positive half-plane the public Lean identity layer uses

`C2.c0_ne_zero_of_re_pos.`

5.2 Transfer principle

The final transfer layer is separate from the construction of F . It uses the identity $F = c_0\zeta$ and nonvanishing of c_0 to transport nonvanishing from the C2 numerator to ζ .

Theorem 5.1 (C2/zeta transfer). *Assume the C2/zeta identity on the off-critical strip and assume that the C2 operator F is nonvanishing there. Then ζ is nonvanishing on the off-critical strip. Consequently, `mathlib's RiemannHypothesis` follows.*

The review-facing Lean declarations are:

```
C2.PeerReview.C2FundamentalIdentityOnRightHalfPlane
C2.PeerReview.C2FundamentalIdentityOnOffCriticalStrip
C2.PeerReview.ZetaNonvanishing_of_C2Nonvanishing
C2.PeerReview.RiemannHypothesis_of_C2Nonvanishing
```

6 Critical branch barrier and transversal tilt

The critical line also appears internally in the C2 branch geometry. The quadratic branch mass is a function of $\sigma = \text{Re}(s)$ alone:

$$\|\mathcal{W}_\infty(s)\|^2 = \frac{2^{1-4\sigma}}{1-2^{-2\sigma}}.$$

The t -dependence disappears because

$$|2^{-ks}|^2 = 2^{-2k\sigma}.$$

Consequently,

$$\|\mathcal{W}_\infty(s)\|^2 < 1 \iff \sigma > \frac{1}{2},$$

and

$$\|\mathcal{W}_\infty(s)\|^2 = 1 \iff \sigma = \frac{1}{2}.$$

The Lean statements are:

```
C2.branchNormSq_closed_form
C2.branchNormSq_barrier
C2.branchNormSqSigmaT_eq_one_on_critical
```

The transversal displacement from the critical line is written

$$s = \frac{1}{2} + \delta + it.$$

Then

$$n^{-s} = n^{-1/2} n^{-it} n^{-\delta}.$$

The C2 tilt is the real factor

$$\text{Tilt}_\delta(n) = n^{-\delta},$$

and its centered bracket is

$$\Delta^2[\text{Tilt}_\delta](c) = (c-1)^{-\delta} + (c+1)^{-\delta} - 2c^{-\delta}.$$

At $\delta = 0$, the tilt is constant and the bracket vanishes. Away from the critical line the bracket has a definite sign in the strip-relevant range. In Lean this is recorded by:

```
C2.bracket_tilt_zero
C2.bracket_tilt_pos
C2.bracket_tilt_neg
C2.bracket_tilt_zero_iff_delta_zero
```

7 Regional cover of the off-critical strip

The C2 proof of off-critical nonvanishing is assembled from three regimes:

$$\text{near-critical-zero} \cup \text{middle-region} \cup \text{edge-region}.$$

Theorem 7.1 (Regional cover assembly). *If the near-critical-zero, middle-region, and edge-region certificates cover the off-critical strip and each certificate proves nonvanishing of the same C2 operator on its region, then that operator is nonvanishing on the whole off-critical strip.*

In Lean this assembly is exposed by:

```
C2.PeerReview.OffCriticalNonvanishing_of_regionalCover.
```

The corresponding terminal route uses:

```
C2.PeerReview.OffCriticalRegionalCover.
```

8 The genuine C2 operator and continuation

The complete operator used in the final route is the genuine C2 operator, exposed in the public facade as

`C2.PeerReview.GenuineC2Operator.`

The continuation data are packaged by

`C2.PeerReview.GenuineC2ContinuationData.`

The role of this layer is to connect the genuine operator to the continued central odd channel and to provide the identity bridge used by the regional cover. The current route obtains near-critical-zero data from continuation via the internal constructor

`C2.GenuineFInfiniteNearAxisData.of_continuation.`

9 Near-critical-zero regime

Near points associated with the critical line, the route uses local punctured nonvanishing. This must be formulated for arbitrary multiplicity. If

$$\rho = \frac{1}{2} + i\gamma$$

is a critical-line zero of multiplicity m_ρ , then the genuine identity gives

$$\partial_\sigma^{m_\rho} F(\rho) = c_0(\rho) \zeta^{(m_\rho)}(\rho) \neq 0.$$

Taylor expansion in the transversal direction yields

$$F(\rho + \delta) = \frac{\delta^{m_\rho}}{m_\rho!} \partial_\sigma^{m_\rho} F(\rho) + R_{m_\rho+1}(\delta),$$

and therefore a local lower bound of the form

$$|F(\rho + \delta)| \geq A_\rho |\delta|^{m_\rho} - B_\rho(r) |\delta|^{m_\rho+1}.$$

Hence there is a positive radius $\delta_\rho^G > 0$ such that

$$0 < |\delta| < \delta_\rho^G \implies F(\rho + \delta) \neq 0.$$

This regime is not an assumption that all zeros of ζ already lie on the critical line. It is one component of the regional cover: it supplies punctured transverse neighborhoods around the critical-line zero data provided by the continuation package. Off-critical points not covered by those neighborhoods are handled by the middle-region certificate and the edge component of the same cover.

The relevant public data are:

`C2.PeerReview.GenuineNearCriticalZeroData`
`C2.PeerReview.GenuineTaylorBoundsFamilyData`
`C2.PeerReview.GenuineLeibnizTaylorBoundsFamilyData`

Theorem 9.1 (Near-critical-zero certificate). *The continuation package for the genuine C2 operator supplies the near-critical-zero certificate required by the off-critical regional cover.*

This statement is represented internally by

`C2.GenuineFInfiniteNearAxisData.of_continuation.`

10 Middle-region resolvent non-cancellation

The middle-region proof is expressed as a dominance argument. A main dyadic contribution is isolated and compared against residual terms coming from tilt, horizontal, cutoff, tail, and truncation estimates.

The vertical depth ratio is

$$q(s) = 2^{-(1+s)}.$$

On the critical line this has the form

$$q(s) = 2^{-3/2} e^{-it \log 2} = r e^{i\theta}, \quad r = 2^{-3/2}.$$

The full vertical tower is governed by the resolvent

$$T_r(\theta) = \frac{1}{1 - r e^{i\theta}},$$

and the four-level prefix is

$$P_r(\theta) = 1 + r e^{i\theta} + r^2 e^{2i\theta} + r^3 e^{3i\theta}.$$

The corresponding sharp lower bounds are

$$\inf_{\theta} |T_r(\theta)| = \frac{1}{1+r}, \quad \inf_{\theta} |P_r(\theta)| = (1-r)(1+r^2).$$

Thus the quartet is not an experimental choice of four terms; it is a natural finite prefix of the vertical resolvent.

The Lean route records the resolvent and quartet interfaces through:

```
C2.geom_resolvent
C2.resolvent_lower_bound
C2.verticalQuartetPrefix
C2.verticalQuartetPrefix_norm_lower_bound_of_offCriticalStrip
```

Definition 10.1 (Dominant four-level block). *The dominant four-level block is the main finite dyadic contribution used as the lower-bound side of the middle-region non-cancellation argument.*

Definition 10.2 (Residual budget). *The residual budget is the collection of upper bounds for all terms not included in the dominant four-level block.*

Theorem 10.3 (Resolvent non-cancellation criterion). *If the dominant four-level block is strictly larger than the residual budget throughout the middle region, then the C2 operator is nonvanishing throughout that region.*

In its finite adjusted form, the criterion has the schematic shape

$$\text{main} - \text{defect} - \text{tail} - \text{residual} > 0.$$

Equivalently, for a decomposition

$$F_X(s) = Q_{4,X}(s) + \mathcal{T}_{4,X}(s) + R_X(s),$$

a sufficient condition is

$$|Q_{4,X}(s)| > |\mathcal{T}_{4,X}(s)| + |R_X(s)|.$$

The paper uses the formal name “resolvent non-cancellation” for this layer. Historical informal labels are not part of the published theorem names.

The public facade exposes this layer through declarations such as

```
C2.PeerReview.DominantFourLevelBlockInequality,
C2.PeerReview.ResidualBudgetDominance,
C2.PeerReview.ResolventNonCancellationCriterion_of_budgetBounds,
C2.PeerReview.AdjustedResolventNonCancellationCriterion_of_residualBudgetBounds.
```

The current canonical middle-region witness is

```
C2.C2CanonicalClosedScaledMiddleLocalData,
```

with the review-facing alias

```
C2.PeerReview.CanonicalClosedScaledMiddleLocalData.
```

10.1 Bound and witness packages

The main middle-region certificate families include:

```
C2ExpandedScalarScaleData
C2ExpandedHorizontalLayerBudget
C2ExpandedSeedScaledBound
C2ExpandedCutoffScaledBound
C2ExpandedQuartetDominance
C2ExpandedScalarMainInequalities
C2ExpandedScalarLocalBulkData
C2CanonicalClosedScaledLocalData
C2CanonicalClosedScaledResidualBudgetLocalData
C2ResolventNonCancellationGenuineCentralLocalData
```

The continued-model middle region used by the public terminal route is named

```
C2.PeerReview.ContinuedModelTerminalMiddleRegion.
```

The full inventory is maintained in `docs/BOUNDS_CERTIFICATES_WITNESSES.md`.

11 Cutoff and residual control

The finite certified operators approximate the genuine operator through an explicit cutoff. If F_X denotes the cutoff model, the residual is

$$R_X(s) = F_\infty(s) - F_X(s).$$

A typical cutoff contribution has the form

$$g_X(n) = e^{-n/X} - 1,$$

so that the residual can be expanded and bounded as a genuine analytic budget. The cutoff is therefore not an unrelated numerical device; it is one of the quantitative terms in the middle-region certificate.

In the public Lean inventory this layer is represented by cutoff and residual packages such as:

```
C2.PeerReview.BalancingSeedExplicitCutoffEstimates
C2.PeerReview.BalancingSeedExplicitCutoffOneLtEstimates
C2.PeerReview.PointwiseOscillatoryResidualUpper
C2.PeerReview.ExponentialPointwiseOscillatoryResidualUpper
```

12 Gluing and edge-region handling

The regional-cover abstraction includes an edge-region certificate. In the current main route, no nontrivial edge construction is required: the selected continued model uses the empty edge certificate

`C2.C2OddTailContinuedBalancingSeedBulkModelEdgeData.empty.`

This is not an omitted case. The edge certificate is empty because the selected near/middle cover already covers the relevant off-critical region in the terminal route. The empty edge component is still recorded formally, so the regional cover has a uniform near/middle/edge interface.

The gluing theorem is the statement that the near-critical-zero region, the middle region, and the edge region cover the off-critical strip, and that the same C2 operator is nonvanishing on each piece. It concludes

$$F(s) \neq 0 \quad (0 < \operatorname{Re}(s) < 1, \operatorname{Re}(s) \neq 1/2).$$

The public Lean wrapper for this assembly is

`C2.PeerReview.OffCriticalNonvanishing_of_regionalCover.`

13 Terminal certificate and endpoint

The terminal certificate is the compact package consumed by the final theorem. Internally it is the structure

`C2.RiemannHypothesisTerminalData.`

It records the calibrated cutoff parameters, horizontal budgets, and the canonical closed/scaled middle local witness. In the public facade it is named

`C2.PeerReview.RiemannHypothesisTerminalCertificate.`

Field or component	Meaning	Lean source
<code>coreCutoff</code>	cutoff schedule for the certified finite model	<code>RiemannHypothesisTerminalData</code>
<code>K, M</code>	calibrated dyadic and truncation parameters	<code>RiemannHypothesisTerminalData</code>
<code>horizontalConstant, horizontalScale, horizontalRatio</code>	horizontal-budget functions used in the canonical closed/scaled middle route	<code>RiemannHypothesisTerminalData</code>
<code>middle.near</code>	canonical near-axis data for the continued C2 model	<code>C2CanonicalClosedScaledMiddleLocalData</code>
<code>middle.edge</code>	edge component; empty in the selected terminal route	<code>C2CanonicalClosedScaledMiddleLocalData</code>
<code>middle.middle_local</code>	pointwise middle-region local witness proving the canonical closed/scaled budget at each covered point	<code>C2CanonicalClosedScaledMiddleLocalData</code>

Theorem 13.1 (Terminal endpoint). *Given a terminal C2 certificate, the official mathlib proposition `RiemannHypothesis` follows.*

The Lean theorem is:

`C2.PeerReview.RiemannHypothesis_of_terminalCertificate.`

Its proof is a wrapper over the internal declaration

`C2.mathlibRiemannHypothesis_of_terminalData.`

The public facade also exposes the canonical constructor from the continued model near-axis data and middle-local data,

```
C2.PeerReview.TerminalCertificate_of_continuedModelMiddleLocal,
```

and the corresponding endpoint

```
C2.PeerReview.RiemannHypothesis_of_continuedModelMiddleLocal.
```

The public facade also records the reduced pointwise target for the same continued-model middle region:

```
C2.PeerReview.TerminalCertificate_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds
C2.PeerReview.RiemannHypothesis_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds.
```

The continuation-based pair `C2.PeerReview.TerminalCertificate_of_continuationAndMiddleLocal` and `C2.PeerReview.RiemannHypothesis_of_continuationAndMiddleLocal` is still available as an audit variant. As of this working revision, these are explicit route-package endpoints rather than a zero-argument theorem obtained by applying a public `terminalCertificate` constant.

14 Lean implementation

The paper-facing Lean entry point is:

```
import LeanC2.PeerReview
```

This facade imports the concrete route and exposes stable names in namespace

```
C2.PeerReview.
```

The formerly monolithic concrete bulk implementation is now split under:

```
LeanC2/Analytic/GenuineBulkConcrete/.
```

The stable import

```
LeanC2.Analytic.GenuineBulkConcrete
```

remains a facade. The smallest staged module containing the canonical terminal certificate route is

```
LeanC2.Analytic.GenuineBulkConcrete.Terminal.
```

14.1 Paper-to-Lean map

Paper-level statement	Lean declaration
Off-critical nonvanishing implies RH	<code>C2.PeerReview.RiemannHypothesis_of_offCriticalNonvanishing</code>
C2/zeta transfer to zeta nonvanishing	<code>C2.PeerReview.ZetaNonvanishing_of_C2Nonvanishing</code>
C2/zeta transfer to RH	<code>C2.PeerReview.RiemannHypothesis_of_C2Nonvanishing</code>
Regional cover assembly	<code>C2.PeerReview.OffCriticalNonvanishing_of_regionalCover</code>
Middle-region budget criterion	<code>C2.PeerReview.ResolventNonCancellationCriterion_of_budgetBounds</code>
Adjusted residual-budget criterion	<code>C2.PeerReview.AdjustedResolventNonCancellationCriterion_of_residualBudgetBounds</code>
Terminal endpoint	<code>C2.PeerReview.RiemannHypothesis_of_terminalCertificate</code>
Canonical terminal assembly	<code>C2.PeerReview.RiemannHypothesis_of_continuedModelMiddleLocal</code>
Residual terminal assembly	<code>C2.PeerReview.TerminalCertificate_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds</code>
Continued-model residual middle endpoint	<code>C2.PeerReview.RiemannHypothesis_of_continuedModelCanonicalClosedScaledMiddleResidualPointwiseBounds</code>

15 Reproducibility

The certified Lean snapshot currently referenced by this draft is release v1.1.0:

https://github.com/thiagomassensini/dyadic_geometry_zeta_fuction/releases/tag/v1.1.0.

The Zenodo archive for this snapshot is:

<https://doi.org/10.5281/zenodo.20401742>.

The paper source should be archived under a fresh tag when the draft is frozen for review.

The Lean verification command is:

```
lake build LeanC2.Analytic.GenuineBulkConcrete LeanC2.PeerReview LeanC2
```

The command is also run by the GitHub Actions workflow `LeanC2CI`.

16 Discussion

The route is intentionally modular. The C2/zeta transfer, the regional cover, the near-critical-zero package, the middle-region non-cancellation mechanism, and the terminal certificate can be audited separately. This modularity is useful both mathematically and formally: the paper can state conceptual theorems, while the Lean development records the exact certificate objects and bound packages consumed by those theorems.

For external review, the normative maps are:

```
docs/MAIN_ROUTE_MAP.md,           docs/THEOREM_MAP_REVIEW.md,
docs/BOUNDS_CERTIFICATES_WITNESSES.md, docs/LEAN_MODULE_LAYOUT.md,
docs/REPRODUCIBILITY.md.
```

A Non-terminal residual route

The fully explicit finite exact-zeta residual route is kept in the Lean source as a documented obstruction rather than as the current terminal target. The formal statements are:

```
c2ExpandedQuartetResidualMargin_lt_scaledVerticalDepthTail_of_offCriticalStrip
c2AnalyticBulkAllowance_sub_reserve_lt_scaledVerticalDepthTail_of_offCriticalStrip
not_c2CanonicalClosedScaledResidualFiniteExactZetaUpper_lt_analyticResidual_of_offCriticalStrip
```

The terminal route therefore uses sharper middle-region witnesses rather than forcing this coarse residual inequality.

B Naming policy

The paper avoids informal discovery labels. The middle-region mechanism is called a resolvent non-cancellation mechanism, the main finite contribution is called the dominant four-level block, and the correction data are called calibrated residual budgets or calibrated residual seeds. The public naming policy is recorded in `docs/FORMAL_NAMING.md`.

C Audit checks and hidden assumptions

The public Lean tree should be checked for unclosed placeholders with:

```
rg "\bsorry\b|\badmit\b|^\s*axiom\b|^\s*constant\b" LeanC2 LeanC2.lean
```

On the current v1.1.0-based draft this command returns no matches in `LeanC2`. This check is separate from the certificate boundary: the terminal theorem is still a theorem over the explicit `RiemannHypothesisTerminalData` interface unless a closed certificate term is supplied and named.

References

- [1] Thiago F. M. Massensini Silva. *Dyadic Geometry of the Zeta Function – Formal Lean Verification*. GitHub repository, release v1.1.0. https://github.com/thiagomassensini/dyadic_geometry_zeta_fuction.
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